HyperEDL: Spectral–Spatial Evidence Deep Learning for Cross-Scene Hyperspectral Image Classification

Yangbo Feng, Xin Yi, Shuhe Wang, Jun Yue, *Member, IEEE*, Shaobo Xia, and Leyuan Fang, *Senior Member, IEEE*

***Abstract*— Cross-scene hyperspectral image (HSI) classification presents significant challenges due to domain shifts, which amplify epistemic uncertainty and lead to substantial performance drops in unseen scenes. While evidence deep learning (EDL) has shown promise in modeling uncertainty, existing methods fall short, as they do not explicitly account for the epistemic uncertainty arising from spatial–spectral feature interactions. To address these challenges, we propose the spectral–spatial evidence deep learning for cross-scene hyperspectral image classification (HyperEDL) framework, which introduces the spatial–spectral multiorder aggregation module (SS-Moga). This module effectively captures and adaptively encodes multiorder contextual interactions from both spatial and spectral perspectives. By combining multiorder contextual encoding with spatial–spectral confidence, our approach fully aggregates multiorder evidence to mitigate epistemic uncertainty arising from knowledge gaps between seen and unseen scenes. Specifically, it uses Dirichlet distribution to capture correlation between spatial–spectral knowledge about different scenes, which can be generalized to unseen scenes. Extensive experiments on three benchmark datasets demonstrate that HyperEDL outperforms state-of-the-art methods, showcasing its effectiveness and strong generalization ability.**

***Index Terms*— Cross-scene hyperspectral image (HSI) classification, evidence deep learning (EDL), multiorder contexts interaction.**

I. Introduction

HYPERSPECTRAL image (HSI) classification [1], [2], [3] aims to use multimodal spatial–spectral information to identify various ground objects in HSI, which are captured by imaging spectrometers. The benefit of this approach is that it offers a detailed spectral signature [4] for each pixel in the image, enabling fine classification of ground objects. Recently, many studies [5] have achieved strong results on HSI classification, which has been widely applied in diverse domains including precision agriculture [6], wetland monitoring [7], and urban planning [8], among several others. However, a significant challenge arises when dealing with real-world scenarios where spectral characteristics can vary drastically across different scenes. For example, in ground objects' classification [9], [10], [11], the spectral signatures of objects can change significantly in HSIs captured under different scenes, as shown in Fig. 1. Traditional HSI classification models are often trained and tested within the same scene struggling to generalize to unseen scenes. To address this problem, cross-scene HSI classification methods have been explored for identifying objects in unseen scenes.

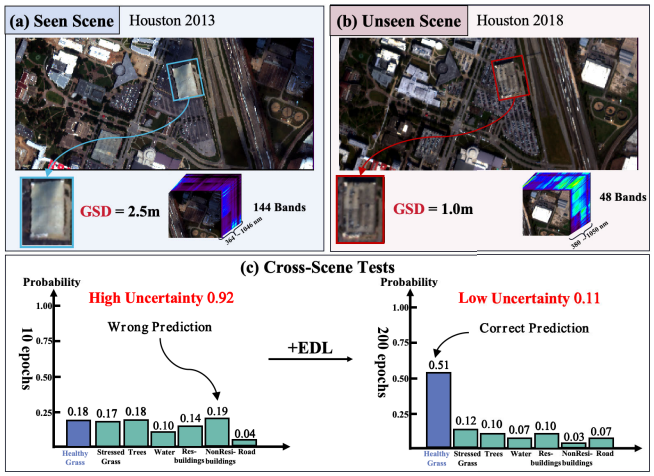


Fig. 1. Uncertainty's role in HSI cross-domain classification. (a) Houston 2013 is the source domain for training. (b) Houston 2018 is the unseen target domain for testing. (c) Bar charts illustrate the effect of cross-domain shifts in the absence of uncertainty modeling. The left chart shows high uncertainty and incorrect predictions, while the right chart highlights how the HyperEDL reduces uncertainty and accurately classifies "Healthy Grass."

The majority of the existing cross-scene HSI classification methods [12], [13], [14], [15], [16], [17] mainly use a generative adversarial learning strategy, which is designed to improve the robustness of the model to the unseen scenes by generate diversified distribution domains. Specifically, single-source domain expansion network [12] is developed to ensure the reliability and effectiveness of scene extension, which applies supervised contrastive learning to learn

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domain-invariant representation. Chen et al. [13] design spectral–spatial adversarial multidomain synthetic network to enhance class-domain semantic consistency information. Tang et al. [14] apply domain adversarial module and biclassifier adversarial module to learn domain-invariant features and task-specific decision boundaries. In addition, to improve the accuracy of the cross-scene HSI classification, a variety of approaches have been adopted, such as spectral–spatial feature enhancement [15] and spectral constraint, language-aware domain adaptation [16], which have achieved impressive performance. The aforementioned methods mostly generate new samples through data augmentation strategies, enabling the model to make more accurate predictions when faced with unseen data, thereby implicitly reducing the model's uncertainty. These methods are still influenced by data, and there are limitations in model's generalization ability for hyperspectral data in unseen scenes.

Despite remarkable progress, the existing cross-scene HSI classification methods still suffer from severe problems that arise from two primary issues as follows.

1) Most strategies for spatial–spectral feature interaction are prone to encoding either extremely low-order [9] or high-order interactions [10] by contextualizing convolutional kernels globally. This impacts the model's generalization to unseen scenes. It is because only extremely low-order feature interaction capture relatively simple and ubiquitous local spatial–spectral concepts, which are frequently inadequate for encapsulating common features or invariant semantic patterns across diverse scenes. In contrast, only high-order feature interaction represents complex concepts of an absolute global scope, which may sacrifice crucial details necessary for cross-scene generalization. Although some works [11], [18] have simultaneously explored the impact of low-order and high-order interactions on models, there are still limitations in the existing middle-order interactions research. The information contained in middle-order features lies between that of low-order and high-order features. They are more complex than low-order features, capable of capturing more details and patterns, yet have not reached the level of high abstraction and generalization characteristic of high-order features. Specifically, from the perspectives of modeling different-order context interaction, these studies have focused on traditional visual recognition tasks, such as traditional image classification [19]. However, they have not fully explored multiorder spatial–spectral feature context interaction within HSIs.
2) The lack of HSIs about unseen scene brings epistemic uncertainty [20], [21] to the learned model. When the epistemic uncertainty is high, it indicates that the model lacks necessary spatial–spectral knowledge to make a confident prediction for encountered cross-scene HSIs. In other words, the model has encountered a sample deviating significantly from existing training data. Thereby, high epistemic uncertainty impedes the model's ability to fully comprehend and effectively generalize to HSI recognition in unseen scenes. Although some works [22], [23] implicitly reduce the epistemic uncertainty of models through data augmentation strategies, there is still a notable lack of research in HSI classification. Specifically, there is little research explicitly focusing on model uncertainty to enhance the model's robustness to unseen cross-scene HSIs.

There are numerous methods for estimating the prediction uncertainty of deep neural networks across various scenarios, including active learning [24], [25] and reinforcement learning [20], [21]. It is not only crucial for safe decision-making in high-risk areas but also indispensable in fields characterized by highly heterogeneous data sources and limited labeled data, especially for remote sensing [26]. Rußwurm et al. [27] apply Bayesian neural network (BNN) to develop model-agnostic methodology for deep recurrent satellite time-series forecasting. Gawlikowski et al. [26] propose a Dirichlet prior network-based model to quantify the distributional uncertainty of deep-learning-based remote sensing models. However, most of the primary techniques for assessing uncertainty in deep learning have centered around BNNs [20], [28] and deep ensembles [29], [30]. Nevertheless, their additional memory and inference time requirements have constrained their viability in real-time and large-scale applications.

These limitations motivate us to design a novel uncertainty estimation architecture for cross-scene HSI classification. The proposed approach, named spectral–spatial evidence deep learning (SS-Evidence Deep Learning), is depicted in Fig. 2 to address the above-mentioned issues.

1) First, a spatial–spectral multiorder aggregation module (SS-Moga) is designed to adaptively extract multiorder contexts, encompassing low-order, middle-order, and high-order feature interactions. This is achieved by contextualizing the convolutional kernel under the guidance of interactions within modern convolutional neural network (CNN), including multiorder spatial–spectral feature aggregation block. Concretely, multiorder spatial aggregation module is applied to combine locality perception with spatial–spectral context aggregation, which can adaptively encode often-neglected inherent different-order context interactions from the spectral channel perspective.
2) Second, to reduce epistemic uncertainty effects, our method formulates it as an uncertainty estimation problem by leveraging evidence deep learning (EDL). HyperEDL uses a deep neural network to predict a Dirichlet distribution of class probabilities, which can be regarded as an different-order context evidence collection process. Different from the traditional Gaussian mixture distribution modeling, Dirichlet distribution can capture correlation between spatial–spectral knowledge about different scenes, which can be generalized to unseen scenes. Moreover, the learned evidences from Dirichlet distribution are informative to quantify the predictive uncertainty of diverse HSI types. For the model with low uncertainty, the reliability of the model is still high even if there is little unseen knowledge in seen scene. This alleviates the problem of knowledge differences in different scenes.

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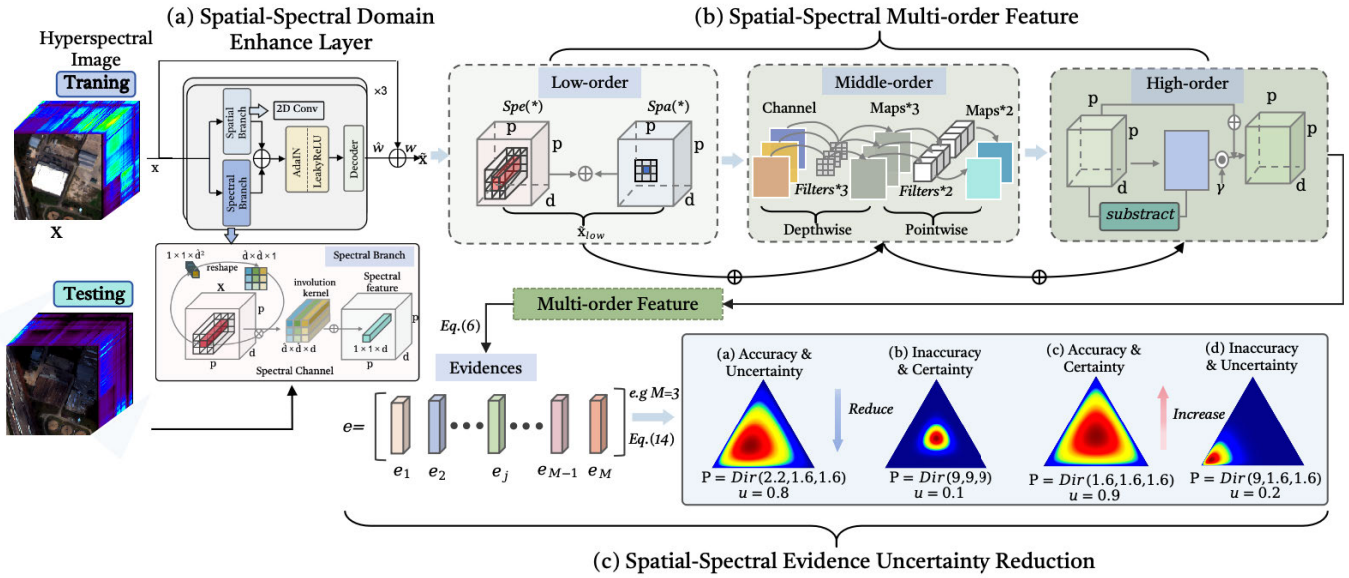


Fig. 2. Overview of the proposed HyperEDL. (a) SSDEL is used to generate enhanced data, which can improve model generalization. (b) Spatial–spectral multiorder aggregation layer is introduced to adaptively encode different-order feature context interactions that would have otherwise been ignored inherently. By carefully designing SS-Moga, the proposed method can adaptively extract multiorder contexts, encompassing low-order, middle-order, and high-order spatial–spectral feature context interactions (c) Example of a three-class classification task for ease of understanding. Specifically, P represents the probability distribution across different categories, and u denotes the uncertainty of the model. After training the model, it is expected that the model will exhibit higher uncertainty when the probability distributions across different categories are similar. Conversely, when the probability distributions across categories differ significantly, the model's uncertainty is expected to be lower. Spatial–spectral evidence uncertainty reduction module is designed to learn evidence from Dirichlet distribution of diverse HSI types. The Dirichlet probability distribution and uncertainty of the data are in agreement. For the model with low uncertainty, the reliability of the model is still high even if there is little unseen knowledge in seen scene. This alleviates the problem of knowledge differences in different scenes.

The contributions of this article are summarized as follows.

1) A novel end-to-end HyperEDL is proposed. To the best of our knowledge, the proposed method is the first attempt to apply EDL to cross-scene HSI classification task, which can reduce the epistemic uncertainty of model.
2) By carefully designing SS-Moga, the proposed method can adaptively extract multiorder contexts, encompassing low-order, middle-order, and high-order spatial–spectral feature context interactions.
3) The proposed framework performs favorably against the state-of-the-art methods on three standard datasets, which verifies its effectiveness and generalization ability.

The rest of this article is organized as follows. We review the related work in Section II. We present the detailed configuration of the proposed framework and the optimization process in Section III. The experimental results are reported and analyzed in Section IV. Finally, conclusions are drawn in Section V.

II. Related Work

This section discusses the existing work closely related to our work regarding cross-scene HSI classification and EDL. The cross-domain hyperspectral classification generally refers to the classification of HSIs obtained from different data sources, different acquisition devices, or under different imaging conditions. These data sources may originate from various satellites, airplanes, or ground sensors, and the environmental conditions during acquisition may also vary. The key to cross-domain classification [31], [32] lies in addressing the differences in data distribution among these various data sources to achieve accurate classification. Different from cross-domain hyperspectral classification, cross-scene hyperspectral classification is conducted in different geographical areas, ecological environments, or application contexts. These scenes may involve diverse vegetation types, soil types, urban structures, and so forth. The challenge of cross-scene classification lies in the potential significant differences in HSI features across different scenes, necessitating models with stronger generalization capabilities.

A. Cross-Scene HSI Classification

Cross-scene HSI classification problem is proposed to improve the robustness of the model in unseen scenes by generating diversified distribution domains, primarily focusing on two aspects. On one hand, the target domain's data are accessible during model training, which belong to the category of semi-supervised domain adaptation. Many approaches [33], [34], [35], [36], [37], [38] have been devised for cross-scene classification from the perspective of semi-supervised domain adaptation, mainly consisting of traditional machine learning and deep learning. Specially, the semi-supervised dual-dictionary nonnegative matrix factorization method [33] is proposed to train two dictionaries for source and target scenes in a low-dimensional subspace, thereby eliminating differences between feature spaces. Liu et al. [34] use a probability-prediction-based maximum mean discrepancy method to investigate classwise adversarial

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adaptation networks, aiming to achieve superior feature-alignment performance. The center-boundary metric loss [39] is designed to learn more discriminative features by considering the distance between sample pairs to enhance intraclass similarity and interclass separability for HSI classification. Mei et al. [40] propose the bidirectional long short-term memory (Bi-LSTM)-based network for HSI classification, which uses a spatial–spectral attention mechanism to emphasize the effective information and reduce the redundant information among spatial–spectral context of pixels. Ma et al. [35] introduce an unsupervised cross-dataset HSI classification method using adversarial domain adaptation to build a discriminator and variational autoencoders as a generator. Wang et al. [36] design hybrid whale optimization algorithm to consider the separability of different land-cover classes and the consistency of selected features between two scenes. Despite certain success achieved by the aforementioned work, the training process necessitates data from the target domain of unseen scenes. However, the difficulty in promptly collecting HSI data for these unseen scenes limits the practical application of the above-mentioned methods.

On the other hand, the target domain's data remain inaccessible, which belong to the category of unsupervised domain adaptation. Recently, numerous single-domain generalization methods have been devised for cross-scene classification, taking into account the constraint that only seen scene data are available for training. Zhang et al. [12] first propose a domain generalization framework for cross-scene HSI classification, which designs a single-source domain expansion network to ensure the reliability and effectiveness of domain extension. A separable deep graph convolutional network [41] integrated with CNN and prototype learning (SDGCP) is applied for HSI classification by extracting effective global structural information. Mei et al. [42] use the spectral super-resolution network (SSR-Net) to learn the transformation model between RGB images and HSIs from training data, and further reduce spectral distortion in the band prediction step. In addition, a language-aware domain generalization network [43] is designed to learn cross-scene-invariant representation with cross-domain shared prior knowledge. Ding et al. [44] develop a consistency-aware customized learning method for cross-scene HSI classification with the joint domain-level and fine-grained class-level distribution alignment. Yang et al. [45] use autoencoders to map input data into a latent space, thereby extracting and preserving the most effective feature representations. However, these approaches only apply high-order semantic information for cross-scene HSI classification. Different from the above deep learning networks, the proposed method not only considers the multiorder spatial–spectra feature information but also reduces the impact of epistemic uncertainty on classification results.

B. Evidence Deep Learning

The assessment of uncertainty in deep neural networks has consistently been a crucial question of importance across various domains, such as evaluating the uncertainty of reinforcement learning models [22] and active learning [18], [23]. Currently, a key major branch within this area is Gaussian processes [19], [46], which excel not only in delivering precise predictions but also in offering reliable assessments of the uncertainty associated with those predictions. Their predictive capabilities have been proven effective across various scenarios, including deep learning [47] and transfer learning [48]. Furthermore, the accuracy of their uncertainty calculations has established them as the leading approach in reinforcement learning models [22] and active learning [23]. Another research branch in prediction uncertainty modeling involves using prior distributions on model parameters and inferring the posterior distribution. Deep ensembles [29], [30] and BNN [20], [28] also fall into this category [49], which leverage prior distributions on synaptic weights to account for parameter uncertainty. However, due to nonlinear activations between layers, calculating the posterior on weights is challenging. Research is ongoing to improve approximation techniques such as variational Bayes [50] and stochastic gradient Hamiltonian Monte Carlo [51] for scalable BNN inference. The uncertainty of a model is mainly classified into aleatoric uncertainty and epistemic uncertainty. Aleatoric uncertainty mainly stems from fluctuations within the data, which are inherent. Conversely, epistemic uncertainty originates from the model's imprecise estimation of input data, and it is influenced by the quality of model training and data. However, the above-mentioned methods face challenges due to their significant memory footprint and inference time demands, which hinder their deployment in real-time industrial-scale environments. To overcome the above limitation, deterministic uncertainty methods (DUMs) have been explored, which can estimate uncertainty in a single forward pass. EDL is one of the most representative research of DUM.

EDL directly applies subjective logic theory and Dempster–Shafer theory of evidence to deep neural networks. Its purpose is to address the inherent weaknesses present in the commonly used softmax classification head. On one hand, since the softmax output essentially represents a point estimate of the probability distribution, it is difficult to leverage it for reliable uncertainty estimation. On the other hand, softmax-based classifiers tend to be overly confident in their incorrect predictions, posing additional challenges to the optimization process. The core concept of EDL involves constructing a parameterized Dirichlet distribution based on this collected evidence and collecting evidence for each category to model the distribution of class probabilities. Besides the probability of each category, the predictive uncertainty can be quantified from the distribution by subjective logic theory in a single forward pass. In addition to providing the probability for each category, this distribution enables the quantification of predictive uncertainty through the use of subjective logic theory.

EDL has been successfully applied to various classification tasks requiring uncertainty modeling. Among them, image classification task is the most representative task for verifying the generalization ability and effectiveness of EDL. Sensoy et al. [52] use EDL to provide better estimates of uncertainty for in-and out-of-distribution samples. After that, a series of EDL approaches are proposed successively. In video

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recognition, Bao et al. [53] apply EDL to distinguish between unknown and known actions in the open-set action recognition task. In the medical field, Gao et al. [54] explicitly apply Dempster–Shafer's evidence theory for staging diagnosis of liver fibrosis with good power of interpretability. Gudhe et al. [55] and Ghesu et al. [56] design multiview evidence deep neural network to extract features from each mammogram's view for screening mammograms assessment. Similarly, in remote sensing, Zhao et al. [57] introduce EDL to the aerial–ground dual-view remote sensing scene classification to model the credibility of each view. Yu et al. [58] propose the evidence graph convolution neural network to enforce the spatial smoothness of the evidence with edge-preserving. Ji et al. [59] combine evidence learning network and generative adversarial network for open-set classification of HSI, which can generate high-quality samples by reducing the distribution discrepancy between generated and real samples. Apart from image classification, EDL is also applied to segmentation and object detection areas. For detection, Park et al. [60] adopt EDL and a new model evidence head, which makes EDL highly compatible with object detection. Su et al. [61] use the EDL to mine the pseudounknown samples from the background and foreground proposal space by the Dirichlet distribution of probability. For segmentation, Ren et al. [62] use EDL to provide pixelwise and image-level confidences, which is reliable medical image analysis. Shi et al. [63] leverage evidence uncertainty estimation for mitochondria segmentation in 3-D electron microscopy images, which can enhance the performance of segmentation. Different from them, our proposed approach preserves multiorder spatial–spectral information based on EDL which could adaptively encode often-neglected inherent different-order interactions from the spectral channel perspective.

III. Proposed Method

This section introduces the pipeline of the end-to-end HyperEDL for cross-scene HSI classification, as shown in Fig. 2. The framework is divided into the spatial–spectral domain enhance layer (SSDEL), spatial–spectral multiorder gated aggregation module, and SS-Evidence Deep Learning layer. First, we give the cross-scene HSI classification formulation and introduce the details of the sample. Then the SSDEL is used to generate enhanced data, which can improve model generalization. In addition, the spatial–spectral multiorder aggregation layer is introduced to adaptively encode different-order feature context interactions that would have otherwise been ignored inherently. Finally, the SS-Evidence Deep Learning layer is designed to learn evidence from Dirichlet distribution of diverse HSI types. For the model with low uncertainty, the reliability of the model is still high even if there is little unseen knowledge in seen scene. This alleviates the problem of knowledge differences in different scenes.

A. Problem Formulation

This article focuses on cross-scene HSI classification. Specifically, seen scene dataset (source domain) and unseen scene dataset (target domain) of HSI with data–label pairs is, respectively, defined as $S = \{(\mathbf{x}_i^s, \mathbf{y}_i^s)\}_{i=1}^M$ and $T = \{(\mathbf{x}_j^t, \mathbf{y}_j^t)\}_{j=1}^N$, where $\mathbf{x}_i^s \in R^{d^s}$ and $\mathbf{x}_j^t \in R^{d^t}$ represent data of the source domain and target domain, respectively, $\mathbf{y}_i^s$ and $\mathbf{y}_j^t$ are the corresponding label. M and N represent the number of samples in the source and target domains, respectively. d^s and d^t are the data dimensions. The data dimensions are the same of the source domain and target domain in this task. This task trains the model in the source domain and tests its generalization ability to unseen scenarios in the target domain. The sample of $p \times p \times d$ spatial patch in HSI is selected from the source domain and sent to the SSDEL for generating enhanced data to improve model generalization. In Sections III-B–III-C, superscript and subscript of $\mathbf{x}_i^s$ are omitted in the partial symbol representation to simplify the symbol representation, and this simplified representation is denoted as $\mathbf{x}$.

B. Spatial–Spectral Domain Enhance Layer

The SSDEL is applied to generate enhanced data, which can improve model generalization. The module structure is shown in Fig. 2. Concretely, considering the rich spatial and spectral information in HSI, spatial branch and spectral branch are designed to extract the above features separately. For the spatial feature learning branch, it includes 2-D convolution layer (2-D Conv) and batch norm operator. The former uses kernels to slide along the spatial dimension and shares convolution kernel weights in the spatial dimension. In the process of learning spatial features, the spectral feature learning branch uses involution convolution kernels that slide along the channel dimension. The shape of these involution convolution kernels is tailored to the shape of the input feature map, making them ideal for extracting spectral features from HSIs. After the spectral feature learning branch and spatial feature learning branch, the spectral feature of the spectral patch is fused with the spatial feature to obtain the spatial–spectral fusion feature $\hat{\mathbf{x}}$. Finally, the spatial–spectral domain enhance data generated by adaptively fusing $\hat{\mathbf{x}}$ with the original spectral patch $\mathbf{x}$ through the learnable weights $\hat{W}$ and W, which can be formalized as

$$\tilde{\mathbf{x}} = W\mathbf{x} + \hat{W}\hat{\mathbf{x}} \tag{1}$$

where the dimension of W and $\hat{W}$ is $\hat{d} \times \hat{d}$, and the dimension of $\tilde{\mathbf{x}}$ is the same as $\mathbf{x}$. To enhance the model's generalization ability for HSIs in unseen scene, both $\mathbf{x}$ and $\tilde{\mathbf{x}}$ are used together to train the model.

C. Spatial–Spectral Multiorder Aggregation Layer

The existing architecture design of deep neural networks frequently lacks hierarchical theory to guide their evolution and modeling, which affect multiorder feature interaction. For instance, CNN uses convolutional and pooling layers to extract local features for low-order interactions. The robustness of CNN stems from their ability to capture numerous low-order interactions, which represent common and widely shared local patterns. However, for high-order feature interaction, despite some networks applying large kernel sizes, they still struggle to encode expressive interactions adequately. This is

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because, while they combine local features to gradually build higher level representations, they are unable to directly encode extremely high-order information interactions. Furthermore, the inappropriate integration of convolutional locality priors with global context injections inherently leads to a bias toward either global shape or local texture, whether achieved through self-attention or convolutions. This results in models being infused with an extreme-order interaction preference ignore middle-order information interaction. Middle-order interactions refer to the interplay between features of medium complexity during the feature extraction process. These features are more intricate than low-order features but have not yet reached a high-order feature of abstraction. When CNN lacks appropriate local priors, they may find it challenging to effectively capture and represent these middle-order features, ultimately leading to a decline in model performance. Therefore, there is a need for a more refined approach that can address these limitations and enhance the ability of CNN to process global contextual information and multiorder interactions.

To tackle these challenges, we introduce the spatial–spectral multiorder aggregation layer (SSMOA) to adaptively encode different-order feature context interactions that would have otherwise been ignored inherently. First, to obtain more effective spatial–spectral information, the spectral feature branch $\mathrm{Spe}(*)$ and the spatial feature branch $\mathrm{Spa}(*)$ are designed to extract the spectral and spatial features separately for any sampled spectral patch. After the spectral feature branch and spatial feature branch, the spectral feature of the spectral patch is fused with the spatial feature to obtain the primary spectral–spatial fusion feature $\tilde{\mathbf{x}}_{\mathrm{pri}}$

$$\tilde{\mathbf{x}}_{\mathrm{pri}} = \hat{W}_1 \mathrm{Spe}\left(\tilde{\mathbf{x}}\right) + \left(1 - \hat{W}_1\right)\mathrm{Spa}\left(\tilde{\mathbf{x}}\right) \tag{2}$$

where $\mathrm{Spe}(*)$ contains a involution operator and a $\mathrm{Conv}_{1\times1}$ convolutional operator. The involution operator focuses on extracting features that are channel-agnostic but spatially specific, thereby enabling the acquisition of richer spectral information. $\mathrm{Spa}(*)$ mainly includes a $\mathrm{Conv}_{3\times3}$ convolutional operator to obtain rich spatial information. $\hat{W}_1$ is an adaptive parameter to balance fused spatial–spectral information.

After that, we use both static and adaptive local perception operations to extract low-order interaction feature $\tilde{\mathbf{x}}_{\mathrm{low}}$

$$\tilde{\mathbf{x}}_{\mathrm{low}} = \mathrm{GeLu}\left(\mathrm{DW}_{3\times3}\left(\mathrm{Conv}_{1\times1}\left(\mathrm{Norm}\left(\tilde{\mathbf{x}}_{\mathrm{pri}}\right)\right)\right)\right) \tag{3}$$

where $\mathrm{Conv}_{1\times1}$ is used to capture fine-grained local spatial features. The $\mathrm{DW}_{3\times3}$ is a depthwise separable convolution operator, which can combine information from different spectral channels through pointwise convolution. It can extract correlated low-order spatial–spectral features, which is crucial for understanding the global information and contextual relationships within HSI. And GeLu and Norm are the activate function and batch norm operation, respectively.

Although low-order features are obtained, to obtain multilevel features, we used residual operation to obtain middle-order feature $\tilde{\mathbf{x}}_{\mathrm{mid}}$. Without additional MLP bottleneck, we introduce a lightweight channel aggregation module to adaptive reallocate channelwise features in high-dimensional hidden spaces and further extend it to a channel aggregation block

$$\tilde{\mathbf{x}}_{\mathrm{mid}} = \tilde{\mathbf{x}}_{\mathrm{low}} + \gamma \odot \left(\tilde{\mathbf{x}}_{\mathrm{low}} - \mathrm{GeLu}\left(\tilde{\mathbf{x}}_{\mathrm{low}} W_r\right)\right) \tag{4}$$

where γ is the channelwise scaling factor initialized as zeros. It reallocates the channelwise feature with the complementary interactions $(\tilde{\mathbf{x}}_{\mathrm{low}} - \mathrm{GeLu}(\tilde{\mathbf{x}}_{\mathrm{low}} W_r))$. The above operation enhances originally overlooked game-theoretic interactions. Furthermore, through residual operation, middle features contain both low-level and mid-level information. Finally, high-order feature is obtained by

$$\tilde{\mathbf{x}}_{\mathrm{hig}} = \sigma\left(\mathrm{Conv}_{1\times1}\left(\tilde{\mathbf{x}}_{\mathrm{mid}}\right) + \tilde{\mathbf{x}}\right) \tag{5}$$

where $\tilde{\mathbf{x}}_{\mathrm{hig}}$ include multiorder information, including low-order, middle-order, and high-order spatial–spectral features of HSI.

D. Spatial–Spectral Evidence Uncertainty Reduction

Inspired by subjective logic theory and Dempster–Shafer theory, EDL is introduced to overcome the shortcomings of softmax-based classifiers. Different from predicting the probability of each class directly, SS-Evidence Deep Learning first gathers evidence $\boldsymbol{e}$ for each class and constructs a Dirichlet distribution of class probabilities. Then, using this Dirichlet distribution, the predictive uncertainty can be quantified through subjective logic theory. To indicate the degree of activation for each class, evidence is defined as a metric that reflects the amount of support gathered from data in favor of assigning a sample to a specific class. EDL aims at predict evidence for each category and build a Dirichlet distribution of class probability. Given a M-class classification problem, let $\boldsymbol{e}$ be the evidence vector predicted for the sample high-order feature $\tilde{\mathbf{x}}_{\mathrm{hig}}$

$$\boldsymbol{e} = f\left(\tilde{\mathbf{x}}_{\mathrm{hig}}; \theta\right) = \left[e_1, \ldots, e_j, \ldots, e_M\right]. \tag{6}$$

Besides, EDL aims at predict evidence for each category and build a Dirichlet distribution of class probability. Given an M-class classification problem, let $\boldsymbol{e}$ be the evidence vector predicted for the sample high-order feature $\tilde{\mathbf{x}}_{\mathrm{hig}}$. e_j means the evidence of the jth class. Besides, subjective logic formalizes the Dempster–Shafer theory of evidence notion of belief assignments over a frame of discernment as a Dirichlet distribution. Hence, it allows one to use the principles of evidential theory to quantify belief masses and uncertainty through a well-defined theoretical framework. More specifically, subjective opinion considers a frame of M mutually exclusive classes by providing a belief mass b_j for each singleton $j = 1, \ldots, M$ and providing an overall uncertainty mass of u. These $M + 1$ mass values are all nonnegative and sum up to one

$$p_j = b_j + \frac{1}{M}u, \quad \sum_{j=1}^{M}(p)_j \tag{7}$$

$$u + \sum_{j=1}^{M} b_j = 1, \quad b_j = \frac{e_j}{S} \text{ and } u = \frac{M}{S}. \tag{8}$$

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Then, the corresponding Dirichlet distribution strength has parameters $\boldsymbol{\alpha} = [\alpha_1, \ldots, \alpha_j, \ldots, \alpha_M]$ with learned evidence vector, and α_j is the evidence for the jth class. Besides, for an M-class setting, the probability mass $\boldsymbol{p} = [p_1, \ldots, p_j, \ldots, p_M]$ is assumed to follow a Dirichlet distribution parameterized by an M-dimensional Dirichlet strength $\boldsymbol{\alpha}$, and p_j is the probability for the jth class. With the Dirichlet assumption, the expectation of the multinomial probability $\boldsymbol{p}$ is given by

$$\mathbf{E}(p_j) = \frac{\alpha_j}{S} = \frac{\alpha_j}{\sum_{j=1}^{M} \alpha_j} = \frac{e_j + 1}{\sum_{j=1}^{M} e_j + M} \tag{9}$$

where $\alpha_j = e_j + 1$. Drawing inspiration from subjective logic theory and Dempster–Shafer theory of evidence, evidential deep learning gathers evidence for each class to parameterize a Dirichlet distribution for class probabilities, allowing for the quantification of predictive uncertainty based on this distribution. The output of a standard neural network classifier is a probability assignment over the possible classes for each sample. However, a Dirichlet distribution parameterized over evidence represents the density of each such probability assignment; hence it models second-order probabilities and uncertainty. In this way, the Dirichlet evidence can be mapped to the subjective opinion by setting following equations. The corresponding Dirichlet distribution $\mathcal{D}(\boldsymbol{p}|\boldsymbol{\alpha})$ is given by $\boldsymbol{\alpha}$

$$\mathcal{D}(\boldsymbol{p}|\boldsymbol{\alpha}) = \begin{cases} \dfrac{1}{B(\boldsymbol{\alpha})} \displaystyle\prod_{j=1}^{M} p_j^{\alpha_j - 1}, & \text{for } \boldsymbol{p} \in S_M \\ 0, & \text{otherwise} \end{cases} \tag{10}$$

$$S_M = \left\{ \mathbf{p} \middle| \sum_{j=1}^{M} p_j = 1 \text{ and } 0 \le p_1, \ldots, p_j \le 1 \right\} \tag{11}$$

where S_M is the M-dimensional unit simplex and $B(\boldsymbol{\alpha})$ is the M-dimensional multinomial beta function. Treating $\mathcal{D}(\boldsymbol{p}|\boldsymbol{\alpha})$ as a prior on the likelihood function $L(\mathbf{y}|\boldsymbol{p})$, we can derive the marginal likelihood function $\mathcal{M}(\mathbf{y}|\boldsymbol{\alpha})$ as follows:

$$\mathcal{M}(\mathbf{y}|\boldsymbol{\alpha}) = \int_{p \in S_M} L(\mathbf{y}|\boldsymbol{p}) \mathcal{D}(\boldsymbol{p}|\boldsymbol{\alpha}) d\boldsymbol{p} \tag{12}$$

where $\mathbf{y}$ is the one-hot label vector. Thereafter, the negative logarithm of the marginal likelihood function, which is the most commonly adopted optimization objective of EDL, can be derived as follows:

$$\mathcal{L}_{\text{edl}} = -\log(\mathcal{M}(\mathbf{y}|\boldsymbol{\alpha})) = \sum_{j=1}^{M} \mathbf{y}_j \left(\log \sum_{j=1}^{M} \alpha_j - \log \alpha_j \right). \tag{13}$$

Note that uncertainty u is inversely proportional to the total evidence. When the total evidence is zero, the uncertainty becomes the maximum, as shown in Fig. 2(c). To enhance the cross-domain generalization ability of the model, we also used the KL divergence loss function for both the augmented data and original data. The pseudocode clarifies the entire process of the method. The entire method training process mainly consists of three steps. The first step is SSDEL. The second step is multiorder feature extraction, including the learning of low-order, mid-order, and high-order feature. The third step is to learn the probability distributions of different classes.

Algorithm 1 HyperEDL

Input: hyperspectral image $\mathbf{x}$
Output: predict label $\mathbf{y}$

// **step 1**: the spatial–spectral domain enhance layer
$\tilde{\mathbf{x}} \leftarrow W\mathbf{x} + \hat{W}\hat{\mathbf{x}}$ // Enhanced Data
// **step 2**: the spatial–spectral multiorder aggregation layer
$\tilde{\mathbf{x}}_{pri} \leftarrow \hat{W}_1 Spe(\tilde{\mathbf{x}}) + (1 - \hat{W}_1) Spa(\tilde{\mathbf{x}})$ // Primary Spectral–spatial Fusion Feature
$\tilde{\mathbf{x}}_{low} \leftarrow GeLu(DW_{3\times3}(Conv_{1\times1}(Norm(\tilde{\mathbf{x}}_{pri}))))$ // Low-order Feature
$\tilde{\mathbf{x}}_{mid} \leftarrow \tilde{\mathbf{x}}_{low} + \gamma \odot (\tilde{\mathbf{x}}_{low} - GeLu(\tilde{\mathbf{x}}_{low} W_r))$ // Middle-order Feature
$\tilde{\mathbf{x}}_{hig} \leftarrow \sigma(Conv_{1\times1}(\tilde{\mathbf{x}}_{mid}) + \tilde{\mathbf{x}})$ // High-order Feature
$\boldsymbol{e} \leftarrow f(\tilde{\mathbf{x}}_{hig}; \theta) = [e_1, \ldots, e_j, \ldots, e_M]$ // Evidence Vector
// **step 3**: the probability distribution
$\mathbf{E}(p_j) \leftarrow \frac{\alpha_j}{S} = \frac{\alpha_j}{\sum_{j=1}^{M} \alpha_j} = \frac{e_j + 1}{\sum_{j=1}^{M} e_j + M}$ // Expectation
$\mathcal{D}(\boldsymbol{p}|\boldsymbol{\alpha}) \leftarrow \begin{cases} \frac{1}{B(\boldsymbol{\alpha})} \prod_{j=1}^{M} p_j^{\alpha_j - 1}, & for\ \boldsymbol{p} \in S_M \\ 0, & otherwise \end{cases}$ // Dirichlet Distribution
$\mathcal{M}(\mathbf{y}|\boldsymbol{\alpha}) \leftarrow \int_{p \in S_M} L(\mathbf{y}|\boldsymbol{p}) \mathcal{D}(\boldsymbol{p}|\boldsymbol{\alpha}) d\boldsymbol{p}$ // Loss Function
return $\mathbf{y}$

IV. Experiments

This section evaluates the proposed method on two widely adopted cross-scene HSI classification benchmarks, including Houston dataset and Pavia dataset. Recent work in this field follows the proposed dataset partitioning protocol presented in [12]. For the Houston dataset, the model is trained on the Houston 2013, and the Houston 2018 is used to test the performance of cross scene HSI classification model. For the Pavia dataset, the model is trained on the University of Pavia (UP) dataset, and the Pavia Center (PC) dataset is applied to test the performance of the model.

A. Datasets and Evaluation Protocol

1) Public Datasets:

a) Houston dataset: The Houston 2013 dataset [64] includes 144 spectral bands, with a wavelength range of 380–1050 nm, and is composed of 349 × 1905 pixels. The image spatial resolution is 2.5 m. In contrast, the Houston 2018 dataset [65] contains 48 spectral bands, also with a wavelength range of 380–1050 nm, and its image spatial resolution is 1.0 m. The two scenes contain seven consistent classes. For the Houston 2013 scene, we extracted 48 spectral bands within the wavelength range of 380–1050 nm, matching those of the Houston 2018 scene, and focused on a common area of 209 × 955 pixels. The classes and the number of samples are listed in Table I. The pseudocolor images of Houston 2013 and Houston 2018 datasets with ground truth are shown in Fig. 3(c) and (d).

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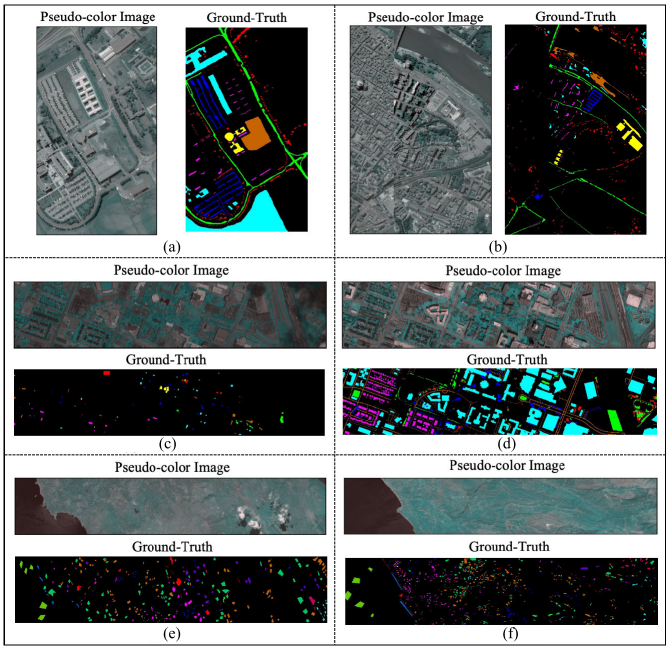


Fig. 3. Pseudocolor images and ground truth for the datasets. (a) Pavia University dataset. (b) PC dataset. (c) Houston 2013 dataset. (d) Houston 2018 dataset. (e) Dioni dataset. (f) Loukia dataset.

TABLE I
CLASSES AND SAMPLE NUMBERS IN THE HOUSTON 2013 AND HOUSTON 2018 DATASETS

Classes		Number of Samples	
No.	Name	Houston 2013	Houston 2018
1	Healthy Grass	345	1353
2	Stressed Grass	365	488
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total		**2530**	**53200**

b) Pavia dataset: The Pavia University dataset is a seen scene dataset, obtained using the reflective optics spectrographic image system. It includes 103 spectral bands, with a wavelength range of 430–860 nm, and is composed of 610 × 340 pixels. The spatial resolution of this dataset is 1.3 m. The PC dataset is an unseen scene dataset, also acquired using the same spectrographic image system. It encompasses 102 spectral bands, with a similar wavelength range of 430–860 nm, and is made up of 1096 × 715 pixels. The spatial resolution of this dataset matches the Pavia University dataset, at 1.3 m. Both the datasets have nine classes of ground objects. However, in this article, we focus on seven identical land cover classes that are common to both the datasets. The class names and sample counts for these seven classes are detailed in Table II. The pseudocolor images of the Pavia University and PC datasets with ground truth are shown in Fig. 3(a) and (b).

c) HyRANK dataset: The dataset encompasses two different regions, which are the Dioni dataset and the Loukia dataset. The Hyperion sensor was used to capture the satellite

TABLE II
CLASSES AND SAMPLE NUMBERS IN THE PAVIA UNIVERSITY AND PC DATASETS

Classes		Number of Samples	
No.	Name	Pavia University	Pavia Center
1	Trees	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1300	7287
5	Shadow	947	2863
6	Meadows	18649	3090
7	Bare soils	5029	6584
Total		**39332**	**39355**

TABLE III
CLASSES AND SAMPLE NUMBERS IN THE DIONI AND LOUKIA DATASETS

Classes		Number of Samples	
No.	Name	Dioni	Loukia
1	Dense Urban Fabric	1262	206
2	Mineral Extraction Sites	204	54
3	Non Irrigated Arable Land	614	426
4	Fruit Trees	105	79
5	Olive Groves	947	1107
6	Coniferous Forest	361	422
7	Dense Sclerophyllous Vegetation	5035	2996
8	Sparce Sclerophyllous Vegetation	6374	2361
9	Sparcely Vegetated Areas	1754	399
10	Rocks and Sand	492	453
11	Water	1612	1393
12	Coastal Water	398	421
Total		**20024**	**10317**

hyperspectral imagery. Regarding the Dioni dataset, it is a seen scene dataset, with dimensions of 250 × 1376 pixels and a spatial resolution of 30 m. The Loukia is an unseen scene dataset, measuring 249 × 945 pixels and sharing the same spatial resolution of 30 m. Both the datasets feature 176 spectral bands and encompass 14 consistent classifications. In this article, we use 12 object categories to assess the model's performance, and the specific sample counts are detailed in Table III. The pseudocolor images of the Pavia University and PC datasets with ground truth are shown in Fig. 3(e) and (f).

2) Evaluation Protocol: To evaluate cross-scene HSI classification model performance from different aspects, the class accuracy (CA), overall accuracy (OA), and kappa coefficient (KC) are considered as evaluation protocol. The CA aims to measure the proportion of correctly classified instances within each individual class. The KC is applied to assess the agreement between the observed classifications and the expected classifications under random chance, providing a measure of the reliability and accuracy of the classifications. These metrics serve as quantitative evaluation tools to rigorously investigate and compare the performance of the proposed method with other comparative methods.

B. Baselines

1) SDENet [12]: The model bases on the idea of domain generalization, which is used to ensure the reliability of domain extension. The method uses generative adversarial learning to train in the source domain and test in the target

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domain. It designs encoder–randomization–decoder to generate the extended domain. In addition, an adversarial strategy is devised to enhance the generator's performance by ensuring that intraclass samples from both the source domain and extended domain are separated.

2) TSTNet [66]: The approach leverages the graph's topological structure alongside a semantic information transfer network to delineate topological relationships. It uses graph optimal transmission to synchronize topological connections between the target and source domains, guided by the principle of maximizing mean difference. Furthermore, a consistency constraint is imposed to merge the outputs of CNN and the graph's topological structure, thereby enhancing the depiction of the feature interplay.

3) LDGNet [43]: The language-aware domain generalization network is proposed to obtain cross-domain-invariant representations leveraging shared prior knowledge across different domains. This technique solely trains on the source domain before transferring the model to the target domain. It uses a dual-stream architecture comprising an image encoder and a text encoder, which facilitates the extraction of visual and linguistic features. In addition, a visual–linguistic alignment is achieved through supervised contrastive learning within this semantic space.

4) LLURNet [67]: In the realm of hyperspectral cross-domain recognition, this study presents a generative extension network. This network comprises a symmetrical encoder–decoder architecture, designed to implicitly establish local joint features while incorporating style randomization. In addition, to mitigate the issue of redundant augmentations, supervised contrastive learning is incorporated. Furthermore, to strike a balance between domain-specific and domain-invariant properties, an adversarial penalty term is formulated. This term leverages interclass and intraclass contrastive regularization within the discriminator.

5) S2ECNet [15]: For hyperspectral cross-domain recognition, this research introduces a generative extension network that implicitly constructs local joint features under randomized styles with a symmetric encoder–decoder architecture. Furthermore, to balance domain-specific and domain-invariant attributes, an adversarial penalty term is formulated through interclass and intraclass contrastive regularization within the discriminator.

6) FDGNet [68]: For hyperspectral cross-domain recognition, this method introduces a frequency disentanglement spectral–spatial encoder. This encoder enables synthesized domains to maintain their semantic consistency while simulating interdomain gaps with the source domain. Furthermore, it integrates data geometry into adversarial training to prevent the creation of unrealistic samples. This approach not only diversifies the new domains but also ensures the data geometry of the extended domains.

7) EHSNet [69]: This domain generalization leverages multilayered high-level semantic info from texts for precise image encoder input. A paradigm aims to extract intricate HSI relationships and land-cover features. In addition, a dual-residual encoder explores HSI spectral–spatial characteristics. By aligning text and image features, EHSNet enhances image encoder representation and achieves zero-shot generalization for cross-scene tasks.

8) S2AMSNet [13]: It is trained on a single-source domain and generalized to unseen domains using spectral–spatial adversarial multidomain synthetic network. Specifically, the spectral–spatial domain-generation network is designed to generate diversified generative domain with the same structure. In addition, a multidomain mixing module is applied to mining class-domain semantic consistency information. Concretely, a semantic consistency discriminator is used to learn cross-domain-invariant knowledge and class-domain semantic consistency information.

C. Implementation Details

The experimental environment is based on the PyTorch framework, the learning rate is set to 0.001, and using the Adam optimizer. The batch size sets on the Houston dataset, Pavia dataset, and HyRANK dataset are 256, 256, and 128, respectively. For the Houston 2013 and Houston 2018 cross-scene HSI classification task, the patch size of $p \times p$ is 13×13. For the Pavia University and PC cross-scene HSI classification task, the patch size of $p \times p$ is 13×13. For the Dioni dataset and the Loukia dataset cross-scene HSI classification task, the patch size of $p \times p$ is 13×13. For the Houston, Pavia and HyRANK datasets, we use 80% as the training set and 20% as the validation set. For the comparison methods, we follow the parameter settings as described in the original article. The dataset is split with 80% for training and 20% for testing. For data augmentation strategies, we follow the S2AMSNet [13]. To evaluate cross-scene HSI classification model performance from different aspects, the CA, OA, and KC are considered for quantitative research. We train model until the loss function is smooth. All the experiments are run on a single NVIDIA GTX 4090 GPU.

D. Performance Comparison

Table IV reports the test accuracy of the cross-scene HSI classification obtained by our model, along with the results of other methods on the Houston 18 dataset based on the same evaluation metrics. In the table, we present the classification accuracy for each class and the average accuracy for all the comparison methods. For the Houston 18 dataset, the SDENet method improves cross-scene HSI classification average accuracy to 76.36% by a single-source domain expansion network. This is because the model is trained using newly generated data, enhancing its generalization ability by increasing the diversity of samples. The LDGNet method performs slightly better than the SDENet method because it uses a dual-stream architecture comprising an image encoder and a text encoder, which facilitates the extraction of visual and linguistic features. In particular, the FDGNet provides a 2% improvement in average accuracy compared with LLURNet on the Houston 18 dataset. This is because frequency disentanglement spectral–spatial encoder enables synthesized domains to maintain their semantic consistency while simulating interdomain gaps with the source domain. However, none of them achieves a substantial improvement. The proposed method

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TABLE IV

CLASSES AND SAMPLE NUMBERS IN THE HOUSTON 2013 AND HOUSTON 2018 DATASETS. HIGHLIGHTING THE BEST RESULTS IN BLACK AND THE SECOND-BEST RESULTS IN BLUE

Classes	Models								
No.	SDENet TIP2023	TSTNet TNNLS2024	LDGNet TGRS2023	LLURNet TGRS2023	S2ECNet TGRS2024	FDGNet TNNLS2024	EHSNet TGRS2024	S2AMSNet TGRS2024	HyperEDL Ours
1	49.59	**85.88**	58.31	**69.48**	48.12	33.19	38.88	48.85	41.98
2	72.05	72.67	81.24	83.92	71.50	70.11	79.54	**85.25**	**86.74**
3	36.15	**64.50**	53.54	62.33	44.79	68.51	55.03	55.10	**69.23**
4	**100.00**	**95.45**	86.36	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	81.82
5	58.00	55.92	79.47	74.60	**81.62**	**83.54**	67.40	70.36	75.86
6	89.19	86.46	**92.37**	83.86	77.61	85.89	87.29	**92.07**	91.85
7	52.76	37.44	14.44	39.31	**70.82**	55.02	35.22	50.90	**57.78**
OA(%)	76.36	75.11	77.84	76.12	74.19	78.27	75.45	**81.32**	**83.25**
KC(%)	58.67	57.83	60.39	60.13	59.16	63.79	58.31	**67.96**	**71.31**

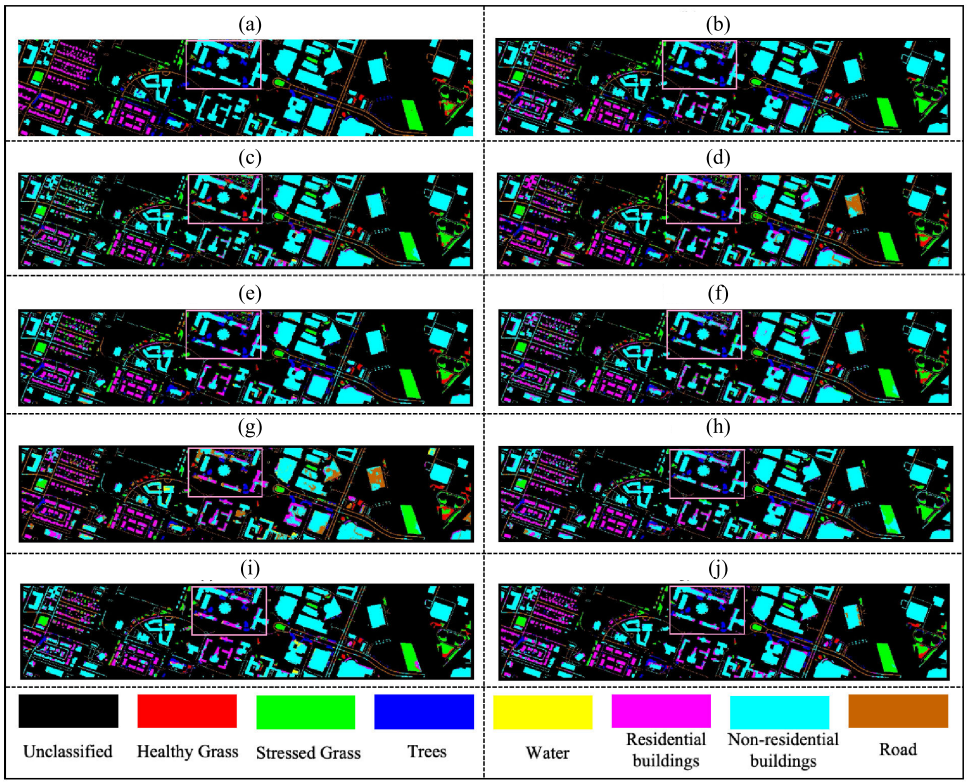


Fig. 4. Visualization and classification maps of different models on the Houston dataset (unseen scene: Houston 2018). (a) Ground truth. (b) HyperEDL. (c) SDENet. (d) TSTNet. (e) LDGNet. (f) LLURNet. (g) S2ECNet. (h) FDGNet. (i) EHSNet. (j) S2AMSNet.

considerably improves the OA and KC. In particular, the OA achieves 83.25%, which exceeds the S2ECNet by 9.06%. The results demonstrate the strong generalize capability to unseen scenes, which can capture correlation between spatial–spectral knowledge about different scenes. In addition, visualization and classification maps of different models on the Houston dataset are shown in Fig. 4.

Tables V and VI report the test results of the cross-scene HSI classification performances obtained by our model, along with the results of other methods on the Pavia dataset and the HyRANK dataset based on the same evaluation metrics. For the Pavia dataset, our method shows better results compared with the other methods on CA, OA, and KC. The OA is used to measure the classification performance of the all the samples, and KC is used to evaluate classification model performance and consistency among evaluators for the same sample. It is obvious that the proposed method shows the best OA of 84.46% and KC 81.34% for the PC dataset. Compared with the TSTNet method, the OA of our method has improved by more than 10%. The major reason is that the TSTNet method ignores the multiorder features of HSI. In addition, HyperEDL method achieves better performance compared with previous FDGNet and S2AMSNet algorithms. This is because the proposed method can reduce epistemic uncertainty effects by leveraging evidential deep learning. EDL is used to predict the Dirichlet distribution of class probabilities, which can be viewed as a process of gathering evidence. The evidence obtained from the Dirichlet distribution is informative for quantifying the predictive uncertainty across various types of HSIs. When the model exhibits low uncertainty, it mitigates the

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TABLE V

CLASSES AND SAMPLE NUMBERS IN THE PAVIA UNIVERSITY AND PC DATASETS. HIGHLIGHTING THE BEST RESULTS IN BLACK AND THE SECOND-BEST RESULTS IN BLUE

Classes	Models								
No.	SDENet TIP2023	TSTNet TNNLS2024	LDGNet TGRS2023	LLURNet TGRS2023	S2ECNet TGRS2024	FDGNet TNNLS2024	EHSNet TGRS2024	S2AMSNet TGRS2024	HyperEDL Ours
1	91.81	68.40	**95.30**	84.14	**96.38**	93.26	95.26	95.06	94.31
2	85.10	85.94	78.99	85.21	**89.29**	75.46	80.82	73.08	**86.49**
3	69.20	62.01	70.17	**76.98**	46.07	73.71	33.04	62.91	**93.40**
4	75.71	53.00	73.69	69.89	84.33	**87.46**	68.27	**87.17**	77.27
5	77.05	92.56	85.96	86.20	**95.35**	90.99	83.44	91.83	**93.50**
6	71.10	**91.55**	74.63	72.27	70.91	60.58	**78.67**	70.03	58.48
7	77.89	78.22	75.62	82.72	64.99	82.53	82.41	**88.03**	**83.23**
OA(%)	80.68	74.45	80.16	80.24	81.73	82.85	78.31	**82.86**	**84.46**
KC(%)	76.72	69.57	76.15	76.30	77.96	**79.42**	73.76	79.38	**81.34**

TABLE VI

CLASSES AND SAMPLE NUMBERS IN THE DIONI AND LOUKIA DATASETS. HIGHLIGHTING THE BEST RESULTS IN BLACK AND THE SECOND-BEST RESULTS IN BLUE

Classes	Models								
No.	SDENet TIP2023	TSTNet TNNLS2024	LDGNet TGRS2023	LLURNet TGRS2023	S2ECNet TGRS2024	FDGNet TNNLS2024	EHSNet TGRS2024	S2AMSNet TGRS2024	HyperEDL Ours
1	36.89	25.24	46.12	**37.38**	34.47	**93.26**	1.46	2.43	25.24
2	0	5.56	**29.63**	0	**83.88**	0	0	0	0
3	29.34	34.51	**67.14**	56.34	63.62	51.41	5.16	0	**67.84**
4	17.72	27.85	**29.11**	8.86	**39.24**	6.33	0	0	5.06
5	0.99	**44.35**	0	0.09	3.34	0.99	0	0	**62.33**
6	**35.34**	7.58	**43.84**	34.12	32.46	31.28	0	0	0
7	72.86	69.89	72.86	76.07	78.81	80.57	**84.58**	**84.01**	67.86
8	**74.38**	**69.34**	64.76	63.73	67.30	68.91	67.60	68.02	62.35
9	70.68	**78.22**	58.00	27.57	48.62	32.58	56.14	69.42	**74.94**
10	0	0.44	0	0.22	0	0.22	**7.28**	**54.53**	2.21
11	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**
12	**100**	**100**	**100**	**100**	**100**	**100**	**92.64**	**100**	**100**
OA(%)	60.96	**63.71**	61.39	59.88	63.49	63.50	60.05	62.67	**64.45**
KC(%)	51.75	**55.67**	52.12	50.41	54.89	52.91	49.67	52.38	**56.42**

issue of knowledge disparities in different scenes. In addition, visualization and classification maps of different models on the Pavia and HyRANK dataset are shown in Figs. 5 and 6.

E. Ablation Experiments

To verify the contribution of each module of the HyperEDL for cross-scene HSI classification, we perform ablation experiments on the relevance of the main modules to understand the modules' significance, including the SSDEL and the SSMOA. In addition, the performance of EDL is verified.

1) Effectiveness of Different Components: We investigate the contributions of the proposed SSDEL and SSMOA to the performance of HyperEDL. We conduct experiments by removing the SSDEL module from our model. The ablation performances of Houston 2018 are shown in Table VII. The results of the Houston dataset are reported, which include OA and KC by our method. For the Houston 2018 dataset, compared with HyperEDL without SSDEL and SSMOA, the performance of OA is increased by 1.64% when the SSDEL is used, and the KC is improved by 1.38%. This is because spatial branch and spectral branch are designed to extract the above rich spatial and spectral information separately. The OA and KC are improved by 2.99% and 4.77%, respectively, when using SSMOA in the model. It is indicated that SSMOA is effective on the cross-scene HSI classification by considering HSI's low-level information, middle-level information, and high-level information. The application of SSDEL and SSMOA provides the best performance with OA 83.25% and KC 71.31% compared with only SSDEL, only SSMOA, and without SSDEL and SSMOA. These results demonstrate the contributions of the SSDEL module and the SSMOA module to performance improvement. The effect of SSMOA in improving OA and KC is greater than that of SSDEL. In addition, we have added the visual images of the Houston dataset and the Pavia dataset both before and after augmentation, as shown in Fig. 7. The patch size of two datasets is 13×13. For the Houston dataset, the bands of hyperspectral data are 48. For the Pavia dataset, the bands of hyperspectral data are 102.

TABLE VII

ABLATION STUDIES FOR THE HOUSTON 2018 DATASET. BEST RESULTS ARE SHOWN IN BOLD

SSDEL	SSMOA	OA(%)	KC(%)
✗	✗	79.95	66.33
✓	✗	81.59	67.71
✗	✓	82.94	71.10
✓	✓	**83.25**	**71.31**

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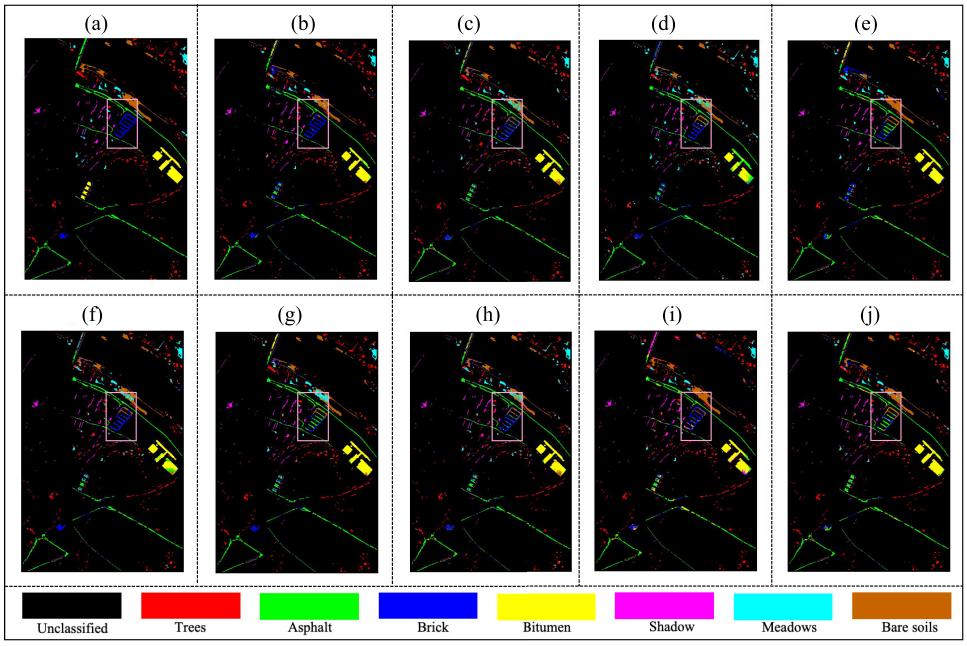


Fig. 5. Visualization and classification maps of different models on the Pavia dataset (unseen scene: PC). (a) Ground truth. (b) HyperEDL. (c) SDENet. (d) TSTNet. (e) LDGNet. (f) LLURNet. (g) S2ECNet. (h) FDGNet. (i) EHSNet. (j) S2AMSNet.

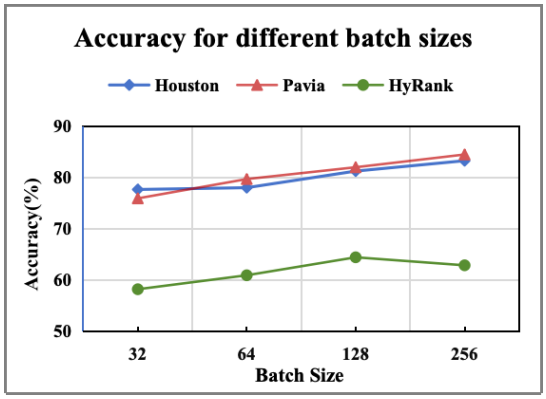


Fig. 6. Experimental accuracy for different datasets with different batch sizes.

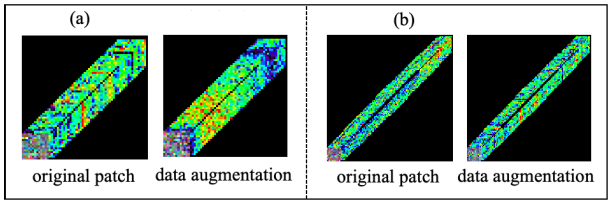


Fig. 7. Visualizing the generated augmented samples. (a) Houston (13×13×48). (b) Pavia (13×13×102).

We have included experimental results for different batch sizes, as shown in Fig. 8. For the Houston 2018 dataset and PC dataset, as shown in Fig. 8, the highest accuracy is achieved when the batch size is 256. For the Loukia dataset of HyRANK, the highest accuracy is obtained with a batch size of 128. For the Loukia dataset, based on our analysis, a smaller batch size generally aids the model in better generalizing to unseen data. This is because a small batch size introduces more randomness during the training process, which helps the model learn more robust feature representations. Conversely, a larger batch size may reduce this randomness, leading to a decline in the model's generalization ability when faced with HSIs of unseen scenes.

TABLE VIII

RESULTS OF SSMOA WITH DIFFERENT-ORDER INFORMATION BEST RESULTS ARE SHOWN IN BOLD

Multi-Order Information				
Low-Order	Middle-Order	High-Order	OA(%)	KC(%)
✗	✗	✗	81.59	67.71
✓	✗	✗	81.78	69.36
✗	✓	✗	82.19	70.17
✗	✗	✓	82.03	69.39
✓	✓	✓	**83.25**	**71.31**

TABLE IX

ABLATION STUDIES FOR THE HOUSTON 2018 DATASET. BEST RESULTS ARE SHOWN IN BOLD

SSMOA	EDL	OA(%)	KC(%)
✗	✗	81.84	67.33
✓	✗	82.03	67.90
✗	✓	82.34	69.93
✓	✓	**83.25**	**71.31**

2) Effectiveness of Multiorder Information: We inspect SSMOA with different-order information, including low-order information, middle-order information, and high-order information. Concretely, Table VIII reports the test results of OA and KC obtained by our algorithm, along with the test results of different level interactive information for SSMOA. We can observe that the performance of cross-scene HSI classification is improved using different-order information. The performance is slightly improved when using the low-order information. It can represent common and widely shared

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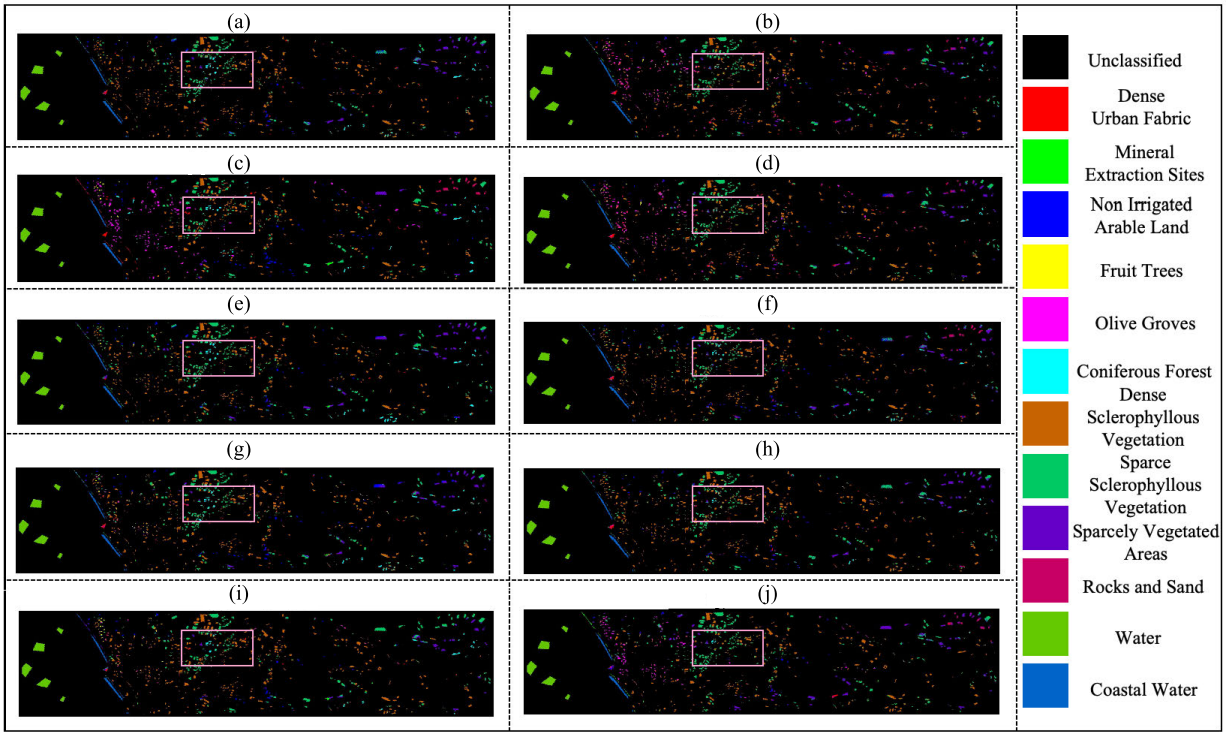


Fig. 8. Visualization and classification maps of different models on the HyRANK dataset (unseen scene: Loukia). (a) Ground Truth. (b) HyperEDL. (c) SDENet. (d) TSTNet. (e) LDGNet. (f) LLURNet. (g) S2ECNet. (h) FDGNet. (i) EHSNet. (j) S2AMSNet.

TABLE X
TRAINING TIME (IN SECONDS) OF DIFFERENT METHODS ON THREE DATASETS

Models	SDENet TIP2023	TSTNet TNNLS2024	LDGNet TGRS2023	LLURNet TGRS2023	S2ECNet TGRS2024	FDGNet TNNLS2024	EHSNet TGRS2024	S2AMSNet TGRS2024	Ours
Houston	18.62	39.23	19.75	7.98	25.76	26.73	19.64	10.93	12.41
Pavia	58.82	35.29	18.46	12.73	49.25	55.78	61.52	20.31	24.32
HyRANK	27.32	34.46	18.10	6.32	22.94	32.63	30.67	12.34	16.16

local patterns. For using middle-order information, we use the primary features as the input for (4). Compared with not using information from any level, using middle-order information improves performance by 0.6%. Middle-order interactions refer to the interplay between features of medium complexity during the feature extraction process. These features are more intricate than low-order features but have not yet reached a high-order feature of abstraction. When information from multiple levels is used simultaneously, the model achieves optimal performance. Information at different levels often contains distinct features and details. Low-order information may focus on basic, localized details, while middle-order and high-level information tends to concentrate on global, more abstract features. By integrating information from these various levels, the model is able to capture more comprehensive and richer data characteristics, thereby enhancing its performance.

3) Effectiveness of EDL: We have researched the effectiveness of the EDL layer with the Dirichlet distribution of the neural network, as shown in Table IX. Concretely, to verify the effectiveness of this module, we removed the EDL layer. The EDL layer is replaced with the cross-entropy loss function. For the Houston 2018 dataset, compared with HyperEDL without the EDL layer and SSMOA, the performance of OA is increased by 0.5% when SSMOA is used, and KC is improved by 2.6%. This is because EDL can rectify model uncertainty by learning Dirichlet distributions for different categories.

Moreover, we evaluate the computational complexity of different methods on three datasets. The training times of all the methods for one epoch are summarized in Table X. All the experiments are conducted in a consistent environment to ensure fairness. As can be seen, although HyperEDL achieves the highest classification accuracy, it has a longer training time compared with the LLURNet method and S2AMSNet method. This is because HyperEDL uses SSMOA with different-order information, including low-order information, middle-order information, and high-order information. Compared with LDGNet, which also uses text information, HyperEDL requires less time, mainly because HyperEDL not uses text features to align with the image features once after entering the semantic space.

V. CONCLUSION

Cross-scene HSI classification is an emerging research domain that possesses adaptability to generalize to unseen scenes across different domains. The existing methods rarely

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consider multiorder spatial–spectral feature interactions, and unseen scenes introduce epistemic uncertainty into the learned model. This article proposes the SS-Evidence Deep Learning to solve the problems in cross-scene HSI classification. The HyperEDL designs SS-Moga to learn multiorder contexts interaction information, which can adaptively encode often-neglected inherent different order interaction. Furthermore, multiorder aggregation evidence is collected to reduce epistemic uncertainty of knowledge differences in seen and unseen scenes. Extensive experimental results demonstrate that our proposed approach performs favorably compared with state-of-the-art methods on three standard datasets, which verifies its effectiveness and generalization ability. Although the proposed method is effective for cross-scene HSI classification tasks, it still encounters difficulties in identifying newly emerging land cover categories in unseen scenarios. In future research, we will develop the model to continuously learn information about new categories.

REFERENCES

[1] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[2] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021.

[3] Q. Liu, J. Yue, Y. Fang, S. Xia, and L. Fang, "HyperMamba: A spectral–spatial adaptive mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5536514.

[4] N. Wambugu et al., "Hyperspectral image classification on insufficient-sample and feature learning using deep neural networks: A review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 105, Dec. 2021, Art. no. 102603.

[5] Y. Liu, J. Yue, S. Xia, P. Ghamisi, W. Xie, and L. Fang, "Diffusion models meet remote sensing: Principles, methods, and perspectives," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4708322.

[6] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-Hi: UAV-borne hyperspectral with high spatial resolution (H2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012.

[7] Y. Huang et al., "Cross-scene wetland mapping on hyperspectral remote sensing images using adversarial domain adaptation network," *ISPRS J. Photogramm. Remote Sens.*, vol. 203, pp. 37–54, Sep. 2023.

[8] B. Yang et al., "Research and application of UAV-based hyperspectral remote sensing for smart city construction," *Cognit. Robot.*, vol. 2, pp. 255–266, Jan. 2022.

[9] L. Agilandeeswari, M. Prabukumar, V. Radhesyam, K. L. N. B. Phaneendra, and A. Farhan, "Crop classification for agricultural applications in hyperspectral remote sensing images," *Appl. Sci.*, vol. 12, no. 3, p. 1670, Feb. 2022.

[10] J. Xie et al., "HyperSFormer: A transformer-based end-to-end hyperspectral image classification method for crop classification," *Remote Sens.*, vol. 15, no. 14, p. 3491, Jul. 2023.

[11] Y. Feng, J. Gao, S. Yang, and C. Xu, "Spatial–temporal exclusive capsule network for open set action recognition," *IEEE Trans. Multimedia*, vol. 25, pp. 1–16, 2023.

[12] Y. Zhang, W. Li, W. Sun, R. Tao, and Q. Du, "Single-source domain expansion network for cross-scene hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1498–1512, 2023.

[13] X. Chen, L. Gao, M. Zhang, C. Chen, and S. Yan, "Spectral–spatial adversarial multidomain synthesis network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5518716.

[14] X. Tang, C. Li, and Y. Peng, "Unsupervised joint adversarial domain adaptation for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022, Art. no. 5536415.

[15] L. Dong, J. Geng, and W. Jiang, "Spectral–spatial enhancement and causal constraint for hyperspectral image cross-scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5507013.

[16] C. Zhao, B. Qin, S. Feng, W. Zhu, L. Zhang, and J. Ren, "An unsupervised domain adaptation method towards multi-level features and decision boundaries for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5546216.

[17] Y. Feng, J. Gao, and C. Xu, "Learning dual-routing capsule graph neural network for few-shot video classification," *IEEE Trans. Multimedia*, vol. 25, pp. 3204–3216, 2022.

[18] L. Guo et al., "Distilling cross-temporal contexts for continuous sign language recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 10771–10780.

[19] L. Guo, W. Xue, B. Liu, K. Zhang, T. Yuan, and D. Metaxas, "Gloss prior guided visual feature learning for continuous sign language recognition," *IEEE Trans. Image Process.*, vol. 33, pp. 3486–3495, 2024.

[20] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian approximation: Representing model uncertainty in deep learning," in *Proc. 33rd Int. Conf. Mach. Learn.*, Jun. 2016, pp. 1050–1059.

[21] W. Huang, J. Zhang, and K. Huang, "Bootstrap estimated uncertainty of the environment model for model-based reinforcement learning," *Proc. AAAI Conf. Artif. Intell.*, vol. 33, no. 1, pp. 3870–3877, Jul. 2019.

[22] C. Szepesvári, *Algorithms for Reinforcement Learning*. Cham, Switzerland: Springer, 2022.

[23] N. Houlsby, F. Huszár, Z. Ghahramani, and M. Lengyel, "Bayesian active learning for classification and preference learning," 2011, *arXiv:1112.5745*.

[24] Y. Gal, R. Islam, and Z. Ghahramani, "Deep Bayesian active learning with image data," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2017, pp. 1183–1192.

[25] K. Chitta, J. M. Alvarez, and A. Lesnikowski, "Large-scale visual active learning with deep probabilistic ensembles," 2018, *arXiv:1811.03575*.

[26] J. Gawlikowski, S. Saha, A. Kruspe, and X. X. Zhu, "An advanced Dirichlet prior network for out-of-distribution detection in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–19, 2022, Art. no. 5616819.

[27] M. Rußwurm, M. Ali, X. X. Zhu, Y. Gal, and M. Körner, "Model and data uncertainty for satellite time series forecasting with deep recurrent models," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Sep. 2020, pp. 7025–7028.

[28] M. Dusenberry et al., "Efficient and scalable Bayesian neural nets with rank-1 factors," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 2782–2792.

[29] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 6405–6416.

[30] Y. Wen, D. Tran, and J. Ba, "BatchEnsemble: An alternative approach to efficient ensemble and lifelong learning," 2020, *arXiv:2002.06715*.

[31] Y. Huang, J. Peng, G. Zhang, W. Sun, N. Chen, and Q. Du, "Adversarial domain adaptation network with calibrated prototype and dynamic instance convolution for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–13, 2024, Art. no. 5514613, doi: 10.1109/TGRS.2024.3387990.

[32] Y. Huang et al., "Two-branch attention adversarial domain adaptation network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540813.

[33] H. Chen, M. Ye, L. Lei, H. Lu, and Y. Qian, "Semisupervised dual-dictionary learning for heterogeneous transfer learning on cross-scene hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3164–3178, 2020.

[34] Z. Liu, L. Ma, and Q. Du, "Class-wise distribution adaptation for unsupervised classification of hyperspectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 508–521, Jan. 2021.

[35] X. Ma, X. Mou, J. Wang, X. Liu, J. Geng, and H. Wang, "Cross-dataset hyperspectral image classification based on adversarial domain adaptation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4179–4190, May 2021.

[36] J. Wang, M. Ye, F. Xiong, and Y. Qian, "Cross-scene hyperspectral feature selection via hybrid whale optimization algorithm with simulated annealing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2473–2483, 2021.

[37] X. Cheng, F. Shi, Y. Zhang, H. Li, X. Liu, and S. Chen, "FRAME: Feature rectification for class imbalance learning," *IEEE Trans. Knowl. Data Eng.*, vol. 37, no. 3, pp. 1167–1181, Mar. 2025.

[38] S. Li et al., "A novel robustness-enhancing adversarial defense approach to AI-powered sea state estimation for autonomous marine vessels," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 55, no. 1, pp. 28–42, Jan. 2025.

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[39] S. Mei, Z. Han, M. Ma, F. Xu, and X. Li, "A novel center-boundary metric loss to learn discriminative features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–16, 2024, Art. no. 5508416, doi: 10.1109/TGRS.2024.3362391.
[40] S. Mei, X. Li, X. Liu, H. Cai, and Q. Du, "Hyperspectral image classification using attention-based bidirectional long short-term memory network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2021.
[41] Y. Lu, S. Mei, F. Xu, M. Ma, and X. Wang, "Separable deep graph convolutional network integrated with CNN and prototype learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.
[42] S. Mei, Y. Geng, J. Hou, and Q. Du, "Learning hyperspectral images from RGB images via a coarse-to-fine CNN," *Sci. China Inf. Sci.*, vol. 65, pp. 1–14, 2022, Art. no. 5501312.
[43] Y. Zhang, M. Zhang, W. Li, S. Wang, and R. Tao, "Language-aware domain generalization network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.
[44] K. Ding, T. Lu, W. Fu, and L. Fang, "Cross-scene hyperspectral image classification with consistency-aware customized learning," *IEEE Trans. Circuits Syst. Video Technol.*, 2024, Art. no. 5526217.
[45] Y. Yang, Y. Xu, Z. Wu, B. Wang, and Z. Wei, "Cross-scene classification of hyperspectral images via generative adversarial network in latent space," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.
[46] D. P. Kingma, T. Salimans, and M. Welling, "Variational dropout and the local reparameterization trick," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, Dec. 2015, pp. 2575–2583.
[47] A. G. Wilson, Z. Hu, R. Salakhutdinov, and E. P. Xing, "Deep kernel learning," in *Proc. Artif. Intell. Statist.*, 2016, pp. 370–378.
[48] M. Kandemir, "Asymmetric transfer learning with deep Gaussian processes," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 730–738.
[49] D. Mackay, "Probable networks and plausible predictions—A review of practical Bayesian methods for supervised neural networks," *Netw., Comput. Neural Syst.*, vol. 6, no. 3, pp. 469–505, Aug. 1995.
[50] C. Blundell, J. Cornebise, K. Kavukcuoglu, and D. Wierstra, "Weight uncertainty in neural network," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 1613–1622.
[51] T. Chen, E. B. Fox, and C. Guestrin, "Stochastic gradient Hamiltonian Monte Carlo," in *Proc. Int. Conf. Mach. Learn. (ICML)*, vol. 32, 2014, pp. 1683–1691.
[52] M. Sensoy, L. Kaplan, F. Cerutti, and M. Saleki, "Uncertainty-aware deep classifiers using generative models," in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2020, pp. 5620–5627.
[53] W. Bao, Q. Yu, and Y. Kong, "Evidential deep learning for open set action recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 13329–13338.
[54] Z. Gao, Y. Liu, F. Wu, N. Shi, Y. Shi, and X. Zhuang, "A reliable and interpretable framework of multi-view learning for liver fibrosis staging," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, Jan. 2023, pp. 178–188.
[55] N. R. Gudhe, M. Sudah, A. Mannermaa, V.-M. Kosma, and H. Behravan, "Multi-view deep evidential fusion neural network for assessment of screening mammograms," Tech. Rep., 2024.
[56] F. C. Ghesu et al., "Quantifying and leveraging classification uncertainty for chest radiograph assessment," in *Proc. 22nd Int. Conf. Med. Image Comput. Comput.-Assist. Intersvent. (MICCAI)*, Shenzhen, China. Cham, Switzerland: Springer, Jan. 2019, pp. 676–684.
[57] K. Zhao, Q. Gao, S. Hao, J. Sun, and L. Zhou, "Credible remote sensing scene classification using evidential fusion on aerial-ground dual-view images," *Remote Sens.*, vol. 15, no. 6, p. 1546, Mar. 2023.
[58] L. Yu, Y. Lou, and F. Chen, "Uncertainty-aware graph-based hyperspectral image classification," in *Proc. The 12th Int. Conf. Learn. Represent.*, 2023, Paper 5503617.
[59] F. Ji et al., "Spectral–spatial evidential learning network for open-set hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.
[60] Y. Park, W. Choi, S. Kim, D.-J. Han, and J. Moon, "Active learning for object detection with evidential deep learning and hierarchical uncertainty aggregation," in *Proc. 11th Int. Conf. Learn. Represent.*, 2023, pp. 1–12.
[61] B. Su, H. Zhang, and Z. Zhou, "HSIC-based moving weight averaging for few-shot open-set object detection," in *Proc. 31st ACM Int. Conf. Multimedia*, Oct. 2023, pp. 5358–5369.
[62] K. Ren et al., "Uncertainty-informed mutual learning for joint medical image classification and segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, Jan. 2023, pp. 35–45.
[63] R. Shi, L. Duan, T. Huang, and T. Jiang, "Evidential uncertainty-guided mitochondria segmentation for 3D EM images," in *Proc. AAAI Conf. Artif. Intell.*, Mar. 2024, vol. 38, no. 5, pp. 4847–4855.
[64] C. Debes et al., "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.
[65] B. Le Saux, N. Yokoya, R. Hansch, and S. Prasad, "2018 IEEE GRSS data fusion contest: Multimodal land use classification [technical committees]," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 1, pp. 52–54, Mar. 2018.
[66] Y. Zhang, W. Li, M. Zhang, Y. Qu, R. Tao, and H. Qi, "Topological structure and semantic information transfer network for cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, 2021, Art. no. 5526512.
[67] H. Zhao, J. Zhang, L. Lin, J. Wang, S. Gao, and Z. Zhang, "Locally linear unbiased randomization network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.
[68] B. Qin, S. Feng, C. Zhao, B. Xi, W. Li, and R. Tao, "FDGNet: Frequency disentanglement and data geometry for domain generalization in cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, 2024, Art. no. 5538314.
[69] X. Wang, S. Dong, X. Zheng, R. Lu, and J. Jia, "Explicit high-level semantic network for domain generalization in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.



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